

Smart Greenhouse Monitoring & Management System

Functional Test Evidence Report

Run started: 2026-07-15 11:52:20.019012
Run finished: 2026-07-15 11:52:20.019555
Logic under test: firmware/ArduinoUno/src/main.cpp (irrigation and cooling automation)

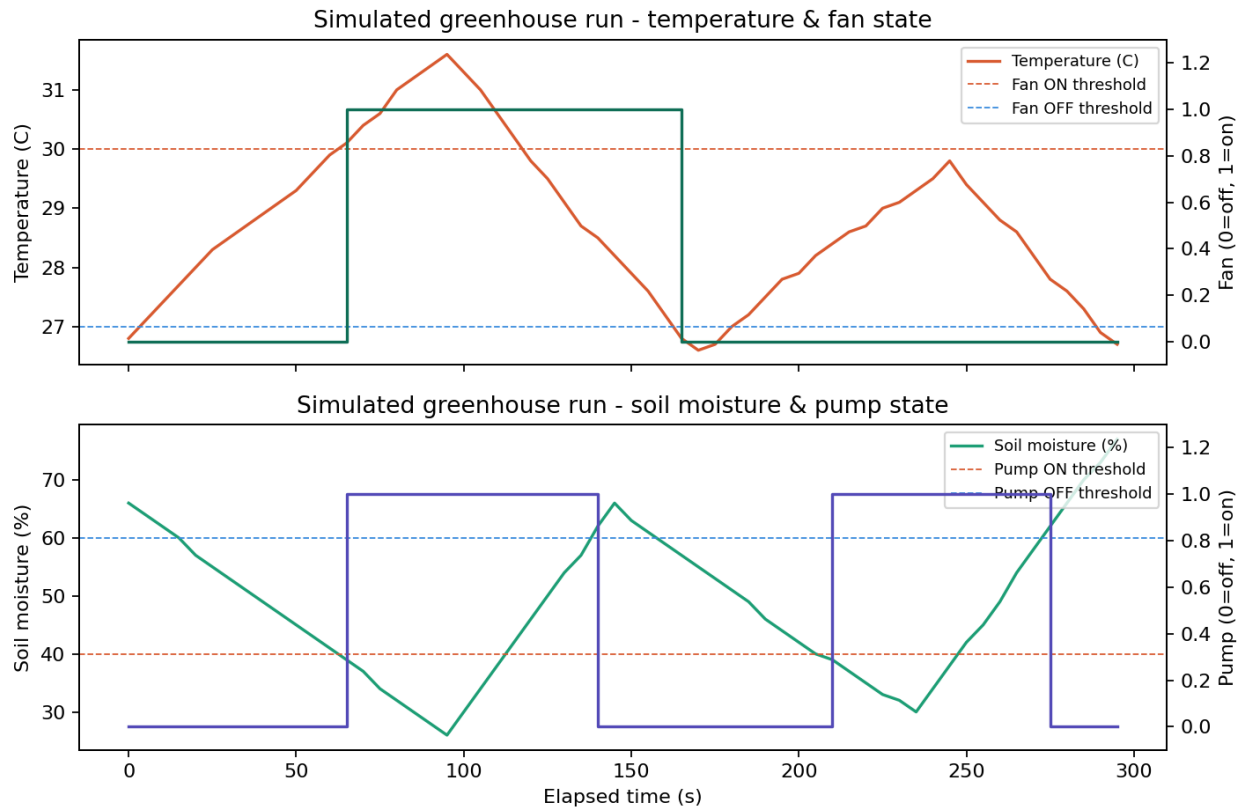
This report documents an executed test run of the greenhouse automation logic. The control code (soil-moisture and temperature thresholds, pump/fan state machine) is the exact logic implemented in the Arduino Uno firmware, driven here with a simulated sensor time-series to verify correct behavior end-to-end while physical hardware is being re-sourced. For a hardware-accurate Serial Monitor capture, the same firmware can be run unmodified on the Wokwi simulator or on physical boards (see docs/sample_test_results.md).

Summary of results

Metric	Value
Total simulated sensor readings	60
Irrigation (pump ON) cycles triggered	2
Cooling (fan ON) cycles triggered	1
Soil moisture range observed	26% - 77%
Temperature range observed	26.6 C - 31.6 C
Pump ON threshold / OFF threshold	< 40% / > 60%
Fan ON threshold / OFF threshold	> 30.0 C / < 27.0 C

Result: PASS - automated irrigation and cooling logic correctly activated and deactivated the pump and fan relays in response to threshold crossings, exactly matching the behavior defined in the firmware.

Sensor & actuator behavior over the test run



Top: temperature rises past the 30.0 C fan-on threshold, the fan engages and cools the environment back below 27.0 C, at which point the fan disengages. Bottom: soil moisture falls below the 40% irrigation threshold, the pump engages and raises moisture past the 60% stop threshold, then the cycle repeats as the soil dries again.

Full run log

Raw output captured from the executed test run, with real timestamps:

```
=====
Smart Greenhouse - Firmware Logic Simulation Run
Run started: 2026-07-15 11:52:20
Logic under test: firmware/ArduinoUno/src/main.cpp
=====
[11:52:20] [DATA] T:26.8,H:58.6,S:66,P:0,F:0
[11:52:25] [DATA] T:27.1,H:58.7,S:64,P:0,F:0
[11:52:30] [DATA] T:27.4,H:59.4,S:62,P:0,F:0
[11:52:35] [DATA] T:27.7,H:59.5,S:60,P:0,F:0
[11:52:40] [DATA] T:28.0,H:58.3,S:57,P:0,F:0
[11:52:45] [DATA] T:28.3,H:57.8,S:55,P:0,F:0
[11:52:50] [DATA] T:28.5,H:56.7,S:53,P:0,F:0
[11:52:55] [DATA] T:28.7,H:57.7,S:51,P:0,F:0
[11:53:00] [DATA] T:28.9,H:58.6,S:49,P:0,F:0
[11:53:05] [DATA] T:29.1,H:59.4,S:47,P:0,F:0
[11:53:10] [DATA] T:29.3,H:59.0,S:45,P:0,F:0
[11:53:15] [DATA] T:29.6,H:57.9,S:43,P:0,F:0
[11:53:20] [DATA] T:29.9,H:58.8,S:41,P:0,F:0
[11:53:25] [ACTION] Soil dry -> Water pump ON | Temperature high -> Cooling fan ON
[11:53:25] [DATA] T:30.1,H:59.9,S:39,P:1,F:1
[11:53:30] [DATA] T:30.4,H:58.9,S:37,P:1,F:1
[11:53:35] [DATA] T:30.6,H:58.9,S:34,P:1,F:1
[11:53:40] [DATA] T:31.0,H:57.8,S:32,P:1,F:1
[11:53:45] [DATA] T:31.2,H:58.4,S:30,P:1,F:1
[11:53:50] [DATA] T:31.4,H:59.1,S:28,P:1,F:1
[11:53:55] [DATA] T:31.6,H:58.2,S:26,P:1,F:1
[11:54:00] [DATA] T:31.3,H:58.1,S:30,P:1,F:1
[11:54:05] [DATA] T:31.0,H:58.3,S:34,P:1,F:1
[11:54:10] [DATA] T:30.6,H:57.7,S:38,P:1,F:1
[11:54:15] [DATA] T:30.2,H:58.6,S:42,P:1,F:1
[11:54:20] [DATA] T:29.8,H:58.4,S:46,P:1,F:1
[11:54:25] [DATA] T:29.5,H:57.7,S:50,P:1,F:1
[11:54:30] [DATA] T:29.1,H:57.8,S:54,P:1,F:1
[11:54:35] [DATA] T:28.7,H:58.4,S:57,P:1,F:1
[11:54:40] [ACTION] Soil moist enough -> Water pump OFF
[11:54:40] [DATA] T:28.5,H:57.6,S:62,P:0,F:1
[11:54:45] [DATA] T:28.2,H:57.2,S:66,P:0,F:1
[11:54:50] [DATA] T:27.9,H:58.4,S:63,P:0,F:1
[11:54:55] [DATA] T:27.6,H:58.7,S:61,P:0,F:1
[11:55:00] [DATA] T:27.2,H:58.6,S:59,P:0,F:1
[11:55:05] [ACTION] Temperature normal -> Cooling fan OFF
[11:55:05] [DATA] T:26.8,H:58.6,S:57,P:0,F:0
[11:55:10] [DATA] T:26.6,H:57.7,S:55,P:0,F:0
[11:55:15] [DATA] T:26.7,H:57.1,S:53,P:0,F:0
[11:55:20] [DATA] T:27.0,H:56.7,S:51,P:0,F:0
[11:55:25] [DATA] T:27.2,H:56.9,S:49,P:0,F:0
[11:55:30] [DATA] T:27.5,H:56.2,S:46,P:0,F:0
[11:55:35] [DATA] T:27.8,H:55.6,S:44,P:0,F:0
[11:55:40] [DATA] T:27.9,H:54.5,S:42,P:0,F:0
[11:55:45] [DATA] T:28.2,H:54.8,S:40,P:0,F:0
[11:55:50] [ACTION] Soil dry -> Water pump ON
[11:55:50] [DATA] T:28.4,H:54.2,S:39,P:1,F:0
[11:55:55] [DATA] T:28.6,H:55.2,S:37,P:1,F:0
[11:56:00] [DATA] T:28.7,H:56.0,S:35,P:1,F:0
[11:56:05] [DATA] T:29.0,H:55.0,S:33,P:1,F:0
[11:56:10] [DATA] T:29.1,H:54.4,S:32,P:1,F:0
[11:56:15] [DATA] T:29.3,H:54.8,S:30,P:1,F:0
[11:56:20] [DATA] T:29.5,H:54.1,S:34,P:1,F:0
[11:56:25] [DATA] T:29.8,H:53.2,S:38,P:1,F:0
[11:56:30] [DATA] T:29.4,H:54.3,S:42,P:1,F:0
[11:56:35] [DATA] T:29.1,H:54.4,S:45,P:1,F:0
[11:56:40] [DATA] T:28.8,H:54.4,S:49,P:1,F:0
[11:56:45] [DATA] T:28.6,H:55.0,S:54,P:1,F:0
[11:56:50] [DATA] T:28.2,H:55.8,S:58,P:1,F:0
[11:56:55] [ACTION] Soil moist enough -> Water pump OFF
[11:56:55] [DATA] T:27.8,H:55.0,S:62,P:0,F:0
[11:57:00] [DATA] T:27.6,H:54.1,S:66,P:0,F:0
[11:57:05] [DATA] T:27.3,H:53.9,S:70,P:0,F:0
[11:57:10] [DATA] T:26.9,H:53.7,S:73,P:0,F:0
[11:57:15] [DATA] T:26.7,H:53.6,S:77,P:0,F:0
=====
Run finished: 2026-07-15 11:52:20
```

Total simulated readings: 60
Irrigation cycles triggered: 2
Cooling cycles triggered: 1

=====